

JORDAN DOYLE

AI Researcher in Automated Software Testing

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Kildare, Ireland



INTRODUCTION

I completed my Ph.D. at the School of Computer Science in University College Dublin in April 2024. Since then, I have been a postdoctoral researcher in the School of Computer Science and Statistics at Trinity College Dublin, where I have also been a member of the Complex Software Lab and SFI Lero. I have also worked in collaboration with ServiceNow as an AI researcher, and Fidelity Investments as a research software engineer. I am looking for a new challenge in the area of software engineering and testing as well as new and exciting opportunities to expand my knowledge and expertise.

EDUCATION

Ph.D. in Computer Science

University College Dublin

September 2017 – January 2024

Dublin, Ireland

I designed, implemented, and evaluated an automated model-based approach to generate test inputs for Android mobile applications from a comprehensive model created using both static and dynamic analysis. I also published and presented three research papers at highly regarded conferences.

Java Python C Bash Android Appium Docker LaTeX



Thesis

Automated Model-based Interface Test Generation for Mobile Applications

B.Sc. in Computer Science (2:1)

University College Dublin

September 2013 – August 2017

Dublin, Ireland

Designed and implemented an end-to-end eReceipt delivery system consisting of a backend database and server, application layer protocol, and a mobile frontend application.

Java Python Ruby C iOS Android SQL Git



Final Year Project

Green Receipt Manager: An eReceipt management and delivery system

PUBLICATIONS



PADRAIG: Precise Android Automated Input Generation. Doyle, J., Laurent, T., Ventresque, A. 24th IEEE International Conference on Software Quality, Reliability, and Security (QRS) 2024



Modelling Android applications through static analysis and systematic exploratory testing. Doyle, J., Laurent, T., Ventresque, A. 10th IEEE International Conference on Dependable Systems and Their Applications (DSA) 2023



Improving Mobile User Interface Testing with Model Driven Monkey Search. Doyle, J., Saber, T., Arcaini, P., Ventresque, A. 15th IEEE Conference on Software Testing, Verification and Validation Workshops (ICSTW) 2021

EXPERIENCE

AI Researcher

Internationalisation @ ServiceNow

August 2024 – Present

Dublin, Ireland

I designed, implemented and evaluated automated solutions capable of detecting internationalisation and localisation defects in the ServiceNow platform. The research involves investigating challenges associated with validating software behaviour across different languages, regions, and cultural settings, including issues related to translated content, user interface adaptation, and regional configuration.

ServiceNow Platform Claude Code Java Selenium Playwright

Research Software Engineer

Research, Development & Innovation @ Fidelity Investments

📅 April 2024 – April 2025

📍 Dublin, Ireland

I developed and deployed an automated solution to detect element reflow defects that occur when accessibility features, such as browser zoom, or text scaling, are used. I extended and adapted existing research tools to integrate with Fidelity's enterprise testing environment and evaluated their effectiveness, usability, and scalability within an industry setting.

Java Selenium WebDriver Selenium Grid Cucumber

Tutor, TA, & Demonstrator

School of Computer Science @ University College Dublin

📅 September 2017 – January 2024

📍 Dublin, Ireland

I assisted the teaching of undergraduate and master's level students in Functional Programming, Operating Systems, and Software Engineering. As a Teaching Assistant (TA), I coordinated teams of up to 10 demonstrators for modules of approximately 200 students. I organised and supervised the grading of student assignments, and module practicals. I was recognised as a distinguished TA for the 2023-24 academic year. As a tutor in the Computer Science Support Centre, I provided one-on-one assistance to students struggling with coursework from any module in Computer Science.

Software Engineering (Java) Operating Systems (Bash) Functional Programming (Scheme)

Test Automation Intern

Movidius @ Intel Ireland

📅 January 2019 – June 2019

📍 Kildare, Ireland

I researched integration testing methods for newly developed Vision Processing Units (VPU's) and their accompanying development API platform. This required the development and configuration of a custom CI/CD pipeline that included both functional and performance tests with direct access to in-development VPU's and dependency hardware.

C CMake GitLab CICD Git

Software Engineering Intern

Insight @ University College Dublin

📅 June 2017 – August 2017

📍 Dublin, Ireland

I extended the features and improved the performance of an Android application. The application was designed to guide recent knee or hip replacement patients through recovery exercises with real time visualisation and feedback using a Unity avatar. The collection of physiotherapy research data for Ph.D. students was also a key part of the project.

Java Android C# Unity Git

Software Engineering Intern

Bruwind @ Vrije Universiteit Brussels

📅 June 2016 – August 2016

📍 Brussels, Belgium

I extended the functionality and features of a web-based application that monitored, analysed and reported the stability of turbine farms in real time, while also collecting data for targeted design optimisations and life-time estimation of turbine substructures.

HTML CSS JavaScript Java JSF PrimeFaces Apache Cassandra Python

Software Engineering Intern

Manufacturing @ Intel Ireland

📅 June 2014/15 – August 2014/15

📍 Kildare, Ireland

I built a prototype sensor platform using an Intel Galileo Gen 2 board and Arduino sensor modules for deployment in Intel's silicon manufacturing environment. The platform collected environmental data for storage in an SQL database and a web application provided real time monitoring and visualisation of the data for the detection of manufacturing faults. I also developed web applications to display, filter, and update data within relational databases.

HTML CSS JavaScript Python ASP VBScript SQL C Arduino SVN Git

SKILLS

Software Engineering Object-Orientated Programming Functional Programming Procedural Programming
Web Development Test Automation Static Analysis Dynamic Analysis Team Coordination & Organisation
Leadership Research & Development Technical & Scientific Writing Presentation & Communication